GIRIDHAR S

Chennai, India | giridharsridharan11@gmail.com | (+91) 915-031-0513 | Giridhar S | GIRIDHAR-VLSI-Engineer
Portfolio

Profile

Final-year Electronics Engineering student specialising in VLSI Design and Embedded Systems, with hands-on experience in physical design automation, GDSII layout, and hardware prototyping. Proven through a state-level hackathon prototype and a VLSI IP-protection tool built with Cadence Virtuoso and KLayout. Actively seeking internship or full-time roles in VLSI/RTL Design or Embedded Systems engineering.

Education

S.A. Engineering College — Chennai, India
B.E. Electronics Engineering — VLSI Design and Technology — *2023 – 2027*

St. Joseph's Matriculation Higher Secondary School — Chennai, India
Higher Secondary Certificate (HSC) — *2023*

St. Joseph's Matriculation Higher Secondary School — Chennai, India
Secondary School Leaving Certificate (SSLC) — *2021*

Technical Skills

Languages	Python, Verilog HDL, C++
EDA Tools	KLayout 0.28, Cadence, Xilinx Vivado, Proteus 8 Professional
VLSI	GDSII Layout, Physical Design, DRC, Dummy Cell Insertion, IP Protection
Libraries	gdspy, pya (KLayout API), Streamlit, NumPy, Pandas, scikit-learn, librosa, Flask
Embedded	Arduino, ESP32, Communication Protocols, RF Communication, Sensor Integration: Ultrasonic (HC-SR04), IR, Proximity, LCD, Servo & Stepper Motors with Motor drivers
Other Tools	MATLAB, PCB Design, VMware Workstation, Ubuntu 24.04 LTS

Projects

Python-Based Layout-Level Dummy Cell Watermarking for IP Protection
Tools: Python, KLayout pya API, Cadence Virtuoso, Assura DRC, GPDK045, Streamlit

- Built an automated VLSI IP-protection tool embedding hidden owner identity as scattered dummy cells in GDSII layouts using BASE64 encoding and SHA-256 cryptographic verification.
- Verified watermark detection on CMOS inverter layouts with zero DRC violations and zero power impact using Cadence Virtuoso + GPDK045 45nm PDK.
- Developed a Streamlit web app supporting real GDS processing, attack simulation, and SHA-256 ownership verification.

Smart Curve Alert System (SCAS) — Hardware Prototype
TN-IMPACK 2026 Hackathon, TANCAME (Govt. of Tamil Nadu) | Team: ASTRIX

- Designed and built a two-module embedded system using Arduino, HC-12 wireless modules, and ultrasonic sensors to detect vehicles at blind curves and deliver real-time alerts.
- Implemented a solar-powered roadside transmitter pole with LED alerts and an in-vehicle receiver with LCD and voice output.
- Presented working physical prototype at **TN-IMPACK 2026**, a state-level hackathon conducted by TANCAME, Government of Tamil Nadu.
- Proposed future integration with EV CAN Bus (ADAS) and V2X/Bluetooth for smart city deployment.

Anti-Deepfake Audio Authentication System
Tools: ESP32, INMP441 I2S Mic, Flask, Python, scikit-learn, librosa, React

- Building a hardware-software system embedding inaudible ultrasonic watermarks at the point of recording using ESP32 with INMP441 I2S microphone.
- Developing Flask/Python backend with cryptographic signing and a scikit-learn/librosa ML model for AI voice detection.
- Integrating React frontend for real-time verification of audio authenticity.

Internships & Training

Intern — 4SG Technology *Aug–Nov 2025 (4 mo)*
Hardware Servicing — Canon, Brother, Epson Copiers

- Serviced and repaired 3–4 Canon, Brother, and Epson copiers daily — covering roller drives, fuser units, toner mechanisms, and gear assemblies.
- Diagnosed and resolved paper feed errors, alignment faults, and mechanical failures, completing most cases on-site without escalation.
- Gained working knowledge of laser and inkjet printing technologies through hands-on repair and preventive maintenance.
- Awarded certificate and Letter of Recommendation for technical competence and professional conduct.

Intern — MaestroMinds — Associate (Project-Based) *Intern → Associate (Ongoing)*
Hardware Projects — Remote Collaboration (Zoho Cliq)

- Contributed to 2 real-time client hardware projects during the internship, working within professional delivery timelines and team workflows.
- Gained hands-on experience with remote collaboration and client communication tools (Zoho Cliq) in a professional project environment.
- Retained as a project-based associate post-internship — assigned to new hardware projects as they arise, with stipend on successful delivery, reflecting consistent performance and trust from the organisation.

Trainee — MSME-DFO Guindy, Govt. of India *Nov 2025–Jan 2026 (6 wk)*
Entrepreneurship Skill Development Programme (ESDP) — IoT Embedded Systems

- Built hands-on proficiency with Arduino Uno R3, Nano, and ESP32 — covering microcontroller architecture, programming, and real-world use cases.
- Worked with core embedded peripherals including LCD displays, servo motors, stepper motors, and potentiometer-based motor control circuits.
- Applied knowledge gained here directly in the Smart Curve Alert System (SCAS) hardware prototype, demonstrating practical continuity from training to project execution.
- Developed foundational understanding of business plan preparation and government MSME schemes for technology entrepreneurs.

Inplant Trainee — Chennai Port Authority *16–21 Jun 2025*
M&E Engineering Department — Electronics & Communication focus

- Completed inplant training across Electronic Data Processing, Electrical, Mechanical, and Electronics divisions, with primary focus on Electronics and Communication systems.
- Observed and studied large-scale industrial communication and electronic systems in an active port infrastructure environment — exposure not accessible in academic settings.
- Adhered to high standards of attendance, conduct, and work performance throughout the training period.

Printing Technician & Counter Staff — Printzo World *Since 2023, part-time*
Print Operations, Customer Handling

- Independently managed print operations serving 150+ customers on peak days, often as the sole operator.
- Operated Canon, Brother, and Epson copiers for bulk and binding orders over 2+ years alongside full-time academics.

Certifications & Workshops

1. Python — MRL Tech Solutions Pvt. Ltd.
2. PCB Design (Hands-on) — Pantech eLearning Pvt. Ltd.
3. Verilog HDL — Pantech eLearning Pvt. Ltd.
4. VLSI Design Flow (RTL to GDS) Workshop — ChipXpert Technologies Pvt. Ltd.